Q-HEALTHGUARD

Privacy-Preserving Quantum Healthcare AI

PROBLEM STATEMENT | VNQFF-02

Hospitals collect sensitive patient information, including medical histories, test results, disease records, laboratory reports, and prescriptions. Because these records are confidential and subject to privacy, security, ethical, and legal constraints, hospitals cannot freely share raw patient data with one another.

As a result, each hospital may train its own AI model using only its local data. This limits collaboration and may restrict how well a model learns across different patient populations. Federated Learning can allow hospitals to train collaboratively without directly exchanging raw records, but model updates can still pose privacy risks unless additional protections are applied.

Main Problem to Solve

> **How can multiple hospitals collaboratively train an effective healthcare AI model using Federated Learning and Quantum Machine Learning, without directly sharing raw patient records, while reducing the risk that exchanged model updates reveal sensitive information?**

Real-World Representation



Team: Epic Engineers • Qiskit Fall Fest 2026 • Proposed project concept; results must be validated experimentally.